

Aritha Munasinghe

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EDUCATION

California State University, Sacramento

Sacramento, CA

Bachelor of Science in Computer Science

Expected December 2027

- **GPA:** 4.0/4.0 (Dean's Honor List) • **Coursework:** Data Structures & Algorithms, System Programming (Unix), Computer Software Engineering, Computer Architecture

EXPERIENCE

Sacramento Municipal Utility District (SMUD)

Sacramento, CA

STEM Student Assistant – Enterprise Applications Engineering

May 2026 – Present

- Built pod-level alerting for non-production **AKS** environments in **Azure Monitor** and **Grafana**, covering sustained CPU (>90% for 10 minutes) and mount-path free space (<10%), replacing a manual check that required shell access into each pod to run **df**.
- Rewrote the initial name-prefix **KQL** query as a controller-aware grouped query, extending a single alert rule from one workload family to every monitored workload group and keeping alerts stable across rolling deployments.
- Shipped **Grafana** configuration changes through direct **Kubernetes** updates after repeated **Helm** release failures, and cleared a stalled rollout by resolving a persistent-volume attachment conflict between the terminating and incoming pods.
- Hardened a scheduled weekly **PowerShell** reporting job across two network environments: gated email delivery on successful export so failures stop reporting silent success, added timestamped per-environment filenames, and 30-day log rotation.

DRM Electronics

Colombo, Sri Lanka

Junior Software Engineer (promoted from Software Engineering Intern)

Aug. 2023 – Aug. 2024

- Optimized **PostgreSQL** schemas and rewrote reporting queries to cut data retrieval latency for high-load analytical reporting modules.
- Refactored legacy **Node.js** backend APIs to raise sustained concurrent request throughput across a multi-tier service architecture.
- Built native Android prototypes in **Java** to validate cross-platform feasibility and user experience.

PROJECTS

KubePrivilegeDiff | *Python, Open Policy Agent (Rego), Kubernetes, NetworkX, GitHub Actions* 2026 – Present

- Modeled Kubernetes **RBAC** as a canonical graph (Subject / Binding / Role / Rule / Scope) using **Pydantic** and **NetworkX**, enabling privilege-path computation across namespace and cluster scopes, with an **OPA/Rego** policy engine holding sole authority over pass/fail verdicts.
- Designed a multi-stage validation gate for AI-generated patches — schema validation, analyzer re-run, **kubectl** dry-run, and ephemeral **kind** cluster apply — such that no unvalidated model output can reach a merge.

Generative AI Knowledge Retrieval Engine | *Python, FastAPI, PostgreSQL (pgvector), RAG* Dec. 2025

- Engineered an end-to-end **RAG (Retrieval-Augmented Generation)** pipeline using **FastAPI** to enable context-aware semantic search over unstructured datasets.
- Implemented **PostgreSQL** with **pgvector** for high-dimensional embedding storage and nearest-neighbour retrieval for LLM grounding.

Real-Time Collaborative Infrastructure | *React, TypeScript, Socket.IO, Docker*

Aug. 2025

- Designed a sandboxed code execution environment in **Docker** with resource limits and no outbound network, supporting multiple language runtimes for concurrent users synchronized over **Socket.IO**.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, C, SQL, PowerShell, Bash, HTML/CSS

Infrastructure & Reliability: Kubernetes (AKS), Docker, Helm, kubectl, Linux/Unix, Azure, Prometheus, Grafana, Azure Monitor, Kusto Query Language (KQL), CI/CD, GitHub Actions, Git

AI/ML & Information Retrieval: RAG, embeddings, pgvector, vector search, LLM output validation, OpenAI API

Security: Open Policy Agent (Rego), policy-as-code, RBAC, threat modeling, static analysis (Semgrep), secret scanning

Web & Data: FastAPI, Node.js, React, Angular, Socket.IO, REST APIs, WebSockets, PostgreSQL

Testing: pytest, mypy, Ruff, Pydantic • **Certification:** Harvard CS50x